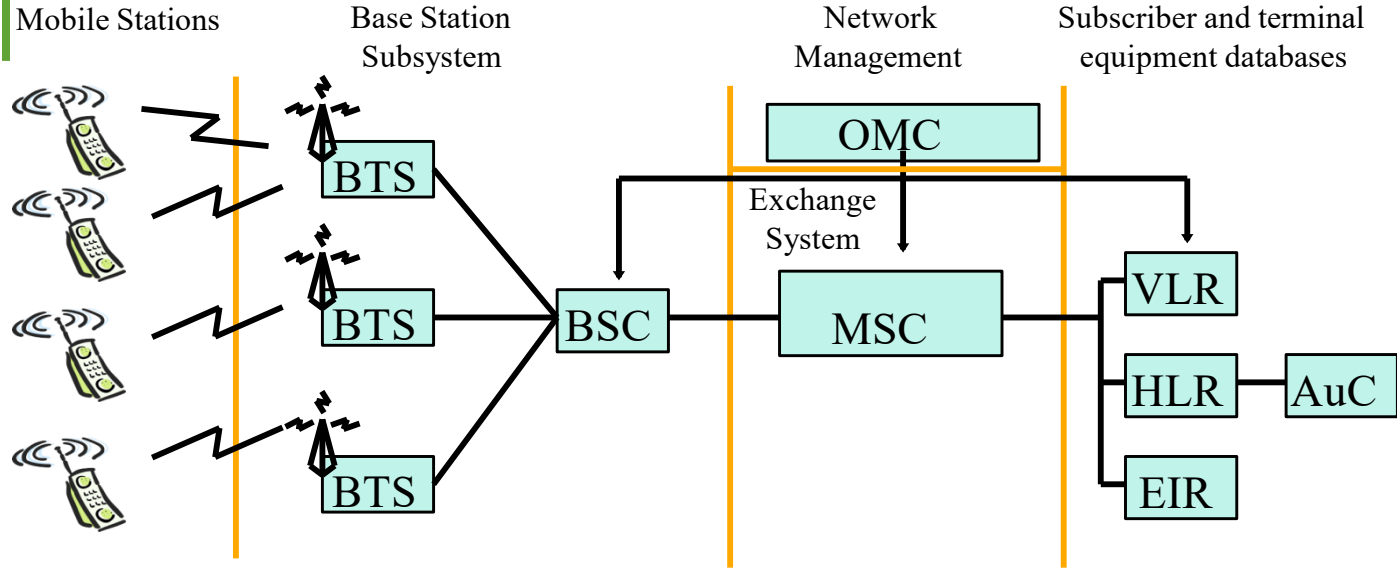


Introduction to Mobile networks

DR. TEWFIQ EL MALIKI

2026

GSM Architecture



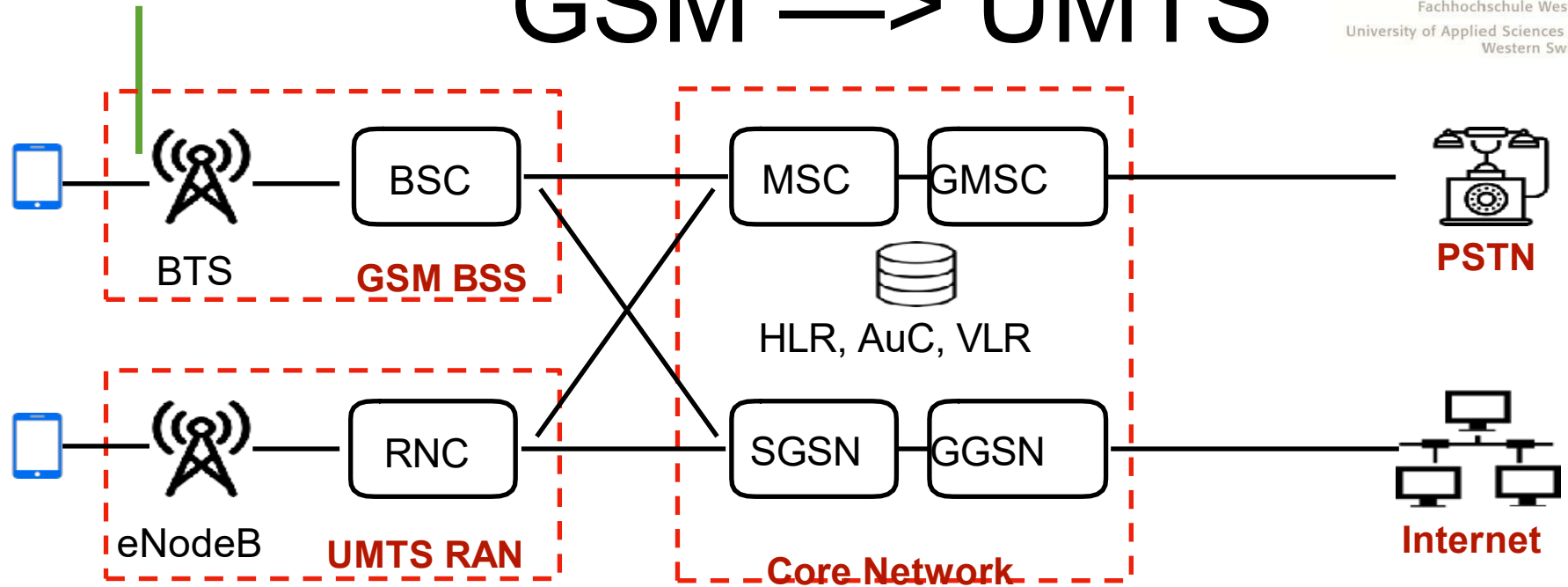
HLR = Home Location Register
AC = Authentication center

VLR = Visitor Location Register
EIR = Equipment Identity Register

GSM Mobile

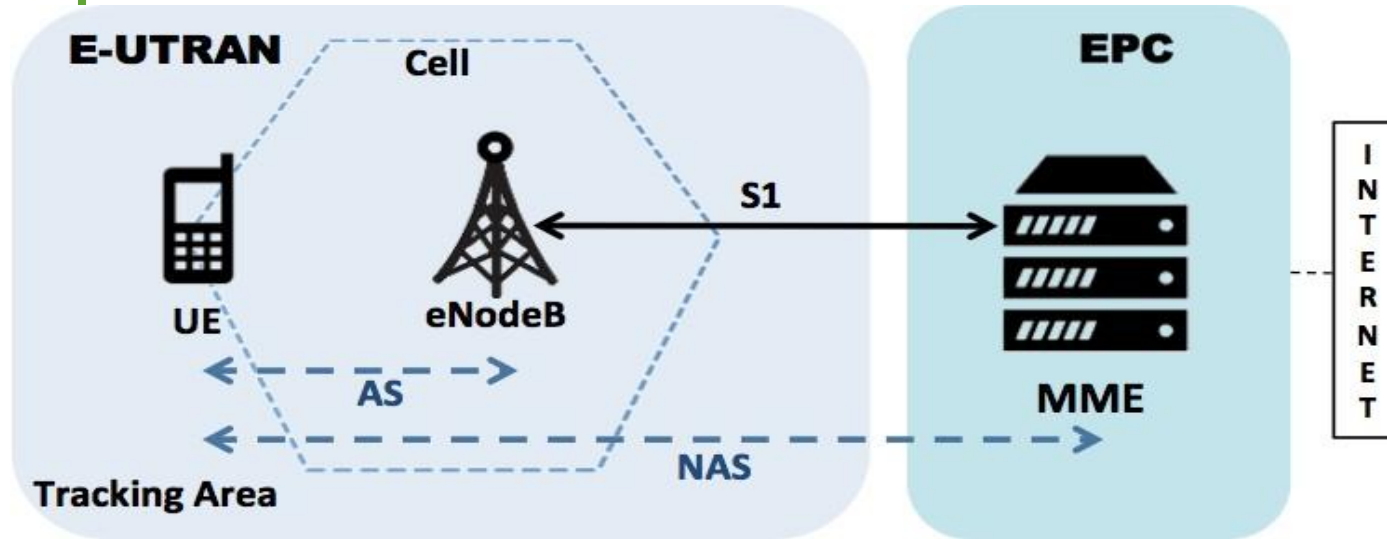
- Mobile Equipment (ME)
- Physical mobile device Identifiers
 - IMEI – International Mobile Equipment Identity
 - Subscriber Identity Module (SIM)
 - Smart Card containing keys, identifiers and algorithms
- Identifiers
 - Ki – Subscriber Authentication Key
 - IMSI – International Mobile Subscriber Identity
 - TMSI – Temporary Mobile Subscriber Identity
 - MSISDN – Mobile Station International Service Digital Network
 - PIN – Personal Identity Number protecting a SIM
 - LAI – location area identity

GSM → UMTS



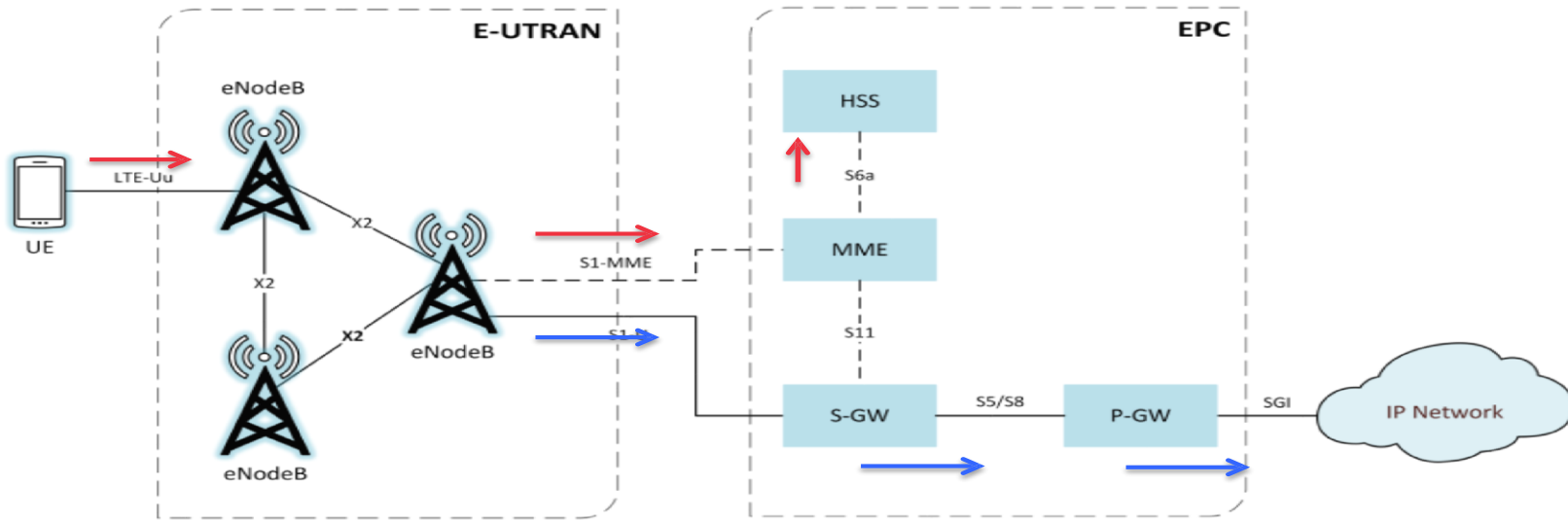
- SIM → USIM
- BTS → eNodeB
- BSC → RNC (Radio Network Controller)
- Backend (MSC, GMSC, SGSN, GGSN all remain the same)

LTE System Architecture



- AS is dedicated to Access messages and NAS for Core messages
- eNodeB: RF component of LTE

LTE System Architecture

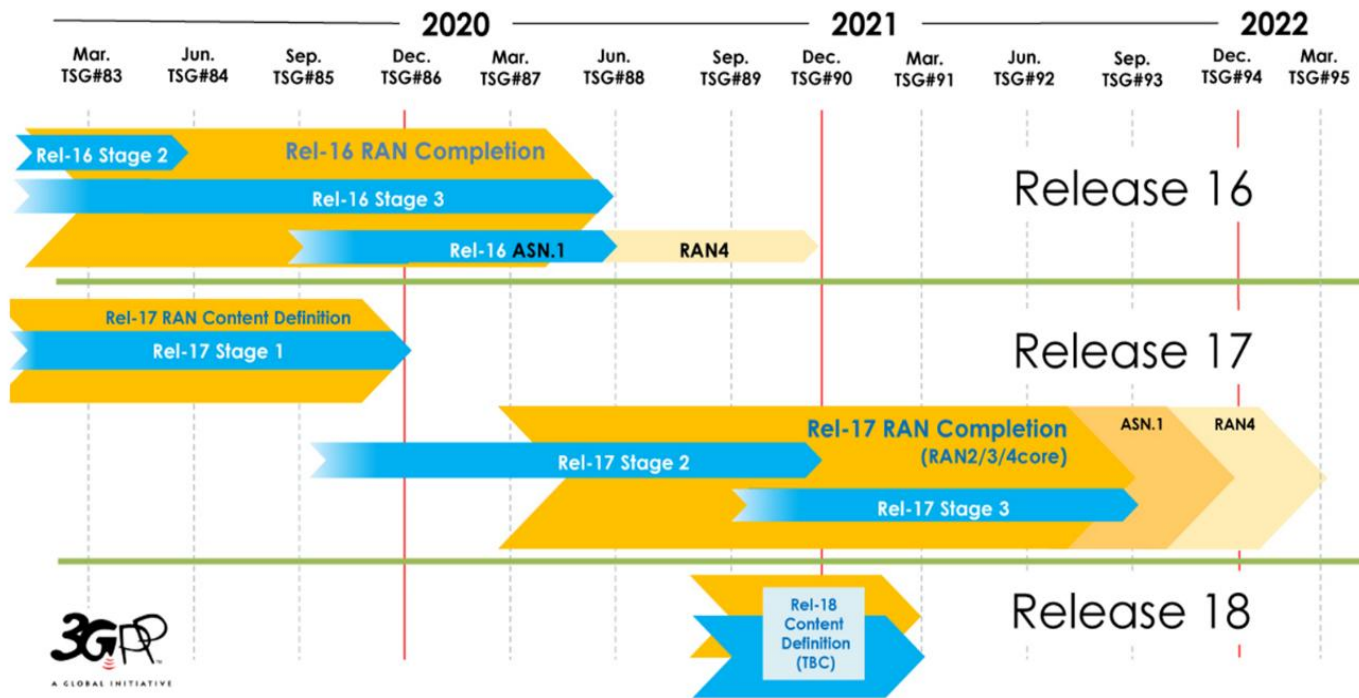


- MME: Mobility Management Entity (Primary Signalling Node)
- HSS: Home subscriber service. Similar to HLR in 2G

Characteristics of LTE Security

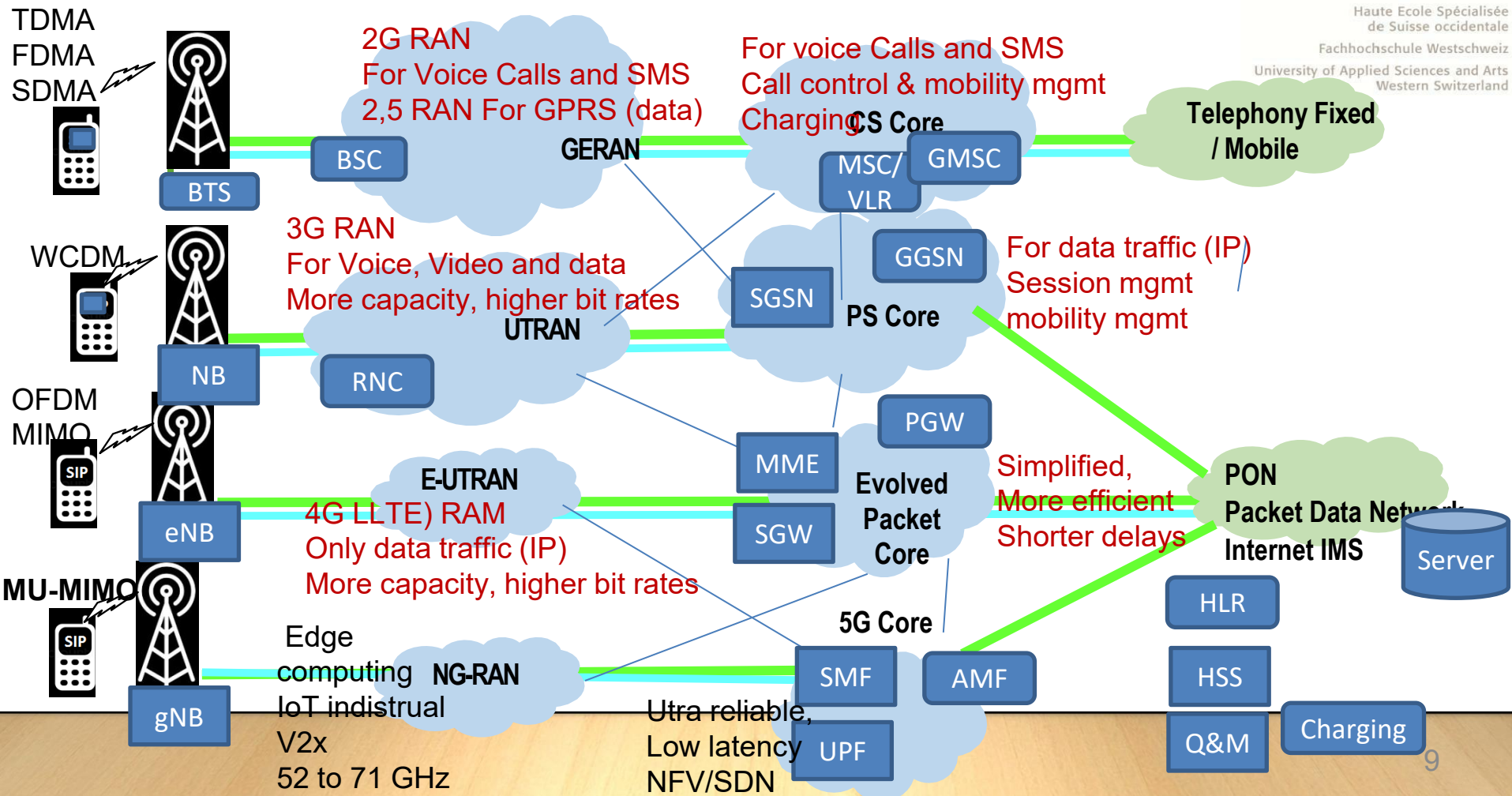
- Re-use of UMTS Authentication and Key Agreement (AKA)
- Extended Key Hierarchy
- Possibility for longer keys
- Greater backhaul protection

Figure 2: 3GPP Release Schedule

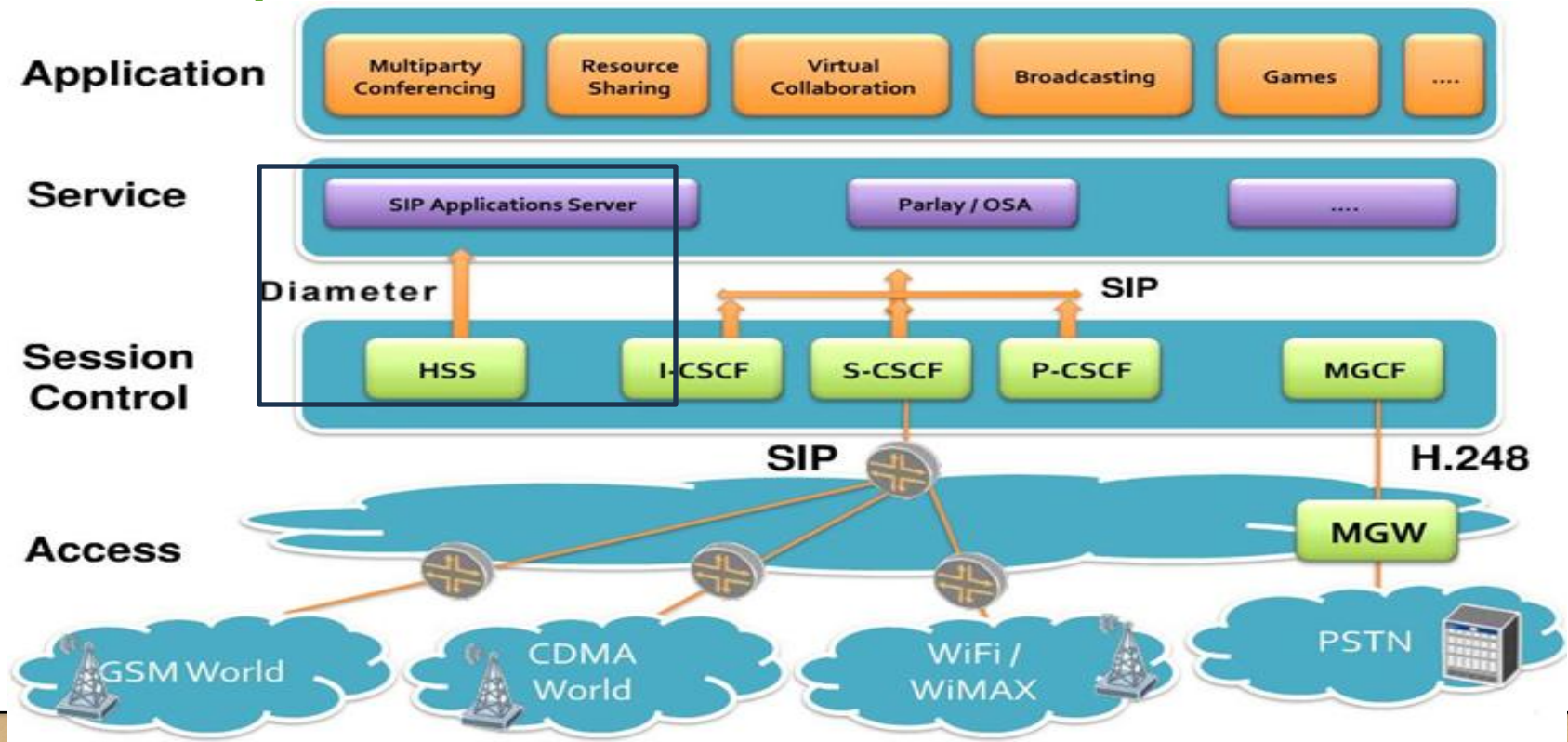


Source: 3GPP TSG SA#87e, 17-20 March 2020, e-meeting document SP-200222

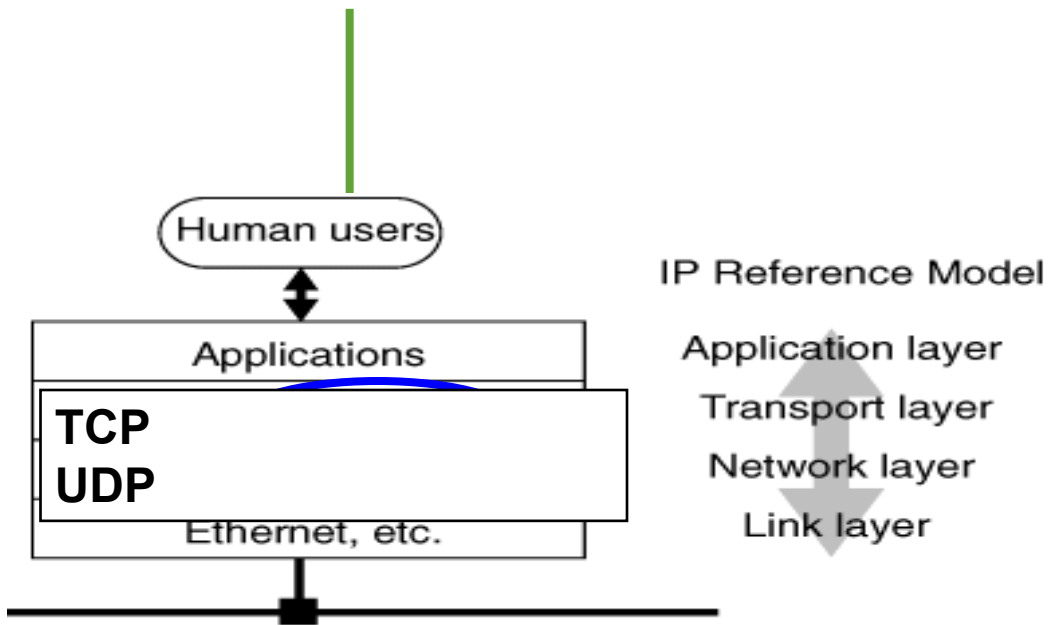
© 3GPP 2020



Architecture réseau IMS



Model Reference



New applications

Migration from PSTN to Packet based Internet

Telephony signaling messages

Shortcomings of existing protocols

TCP

"head-of-line blocking"

Byte-oriented, not message-oriented

**Multi-homing support not built in
DoS attack prone**

UDP

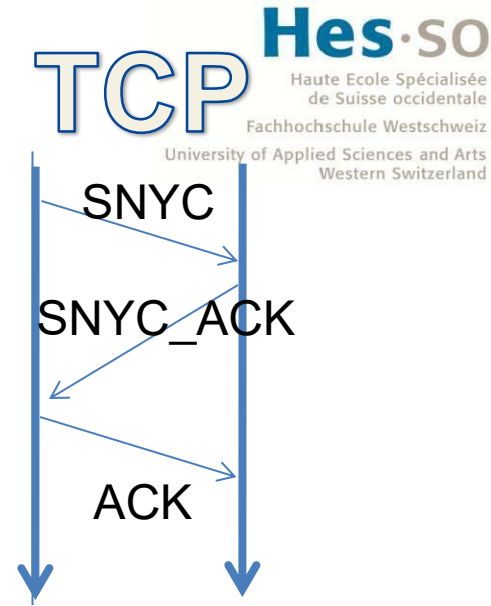
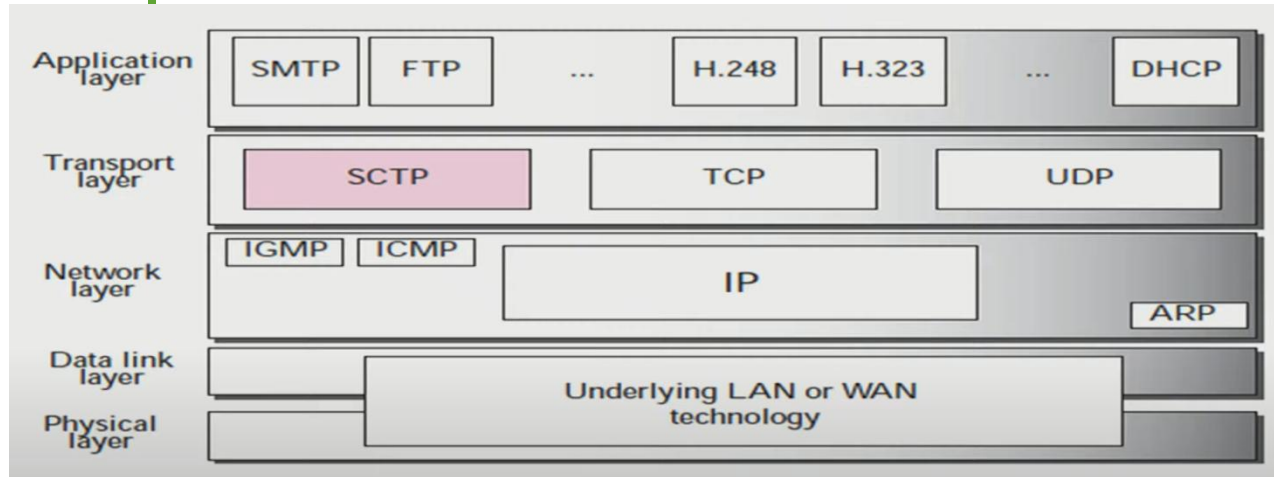
No Reliability

Absence of congestion control

Absence of flow control

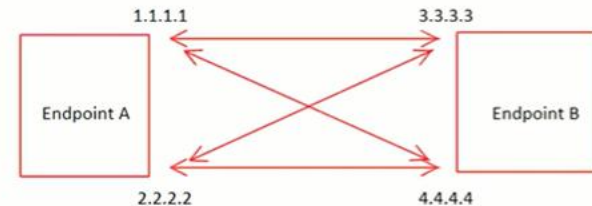
- ✓ “SCTP is a reliable transport protocol operating on top of a connectionless packet network such as IP. ...” – RFC 2960
- ✓ Has built-in support for multi-homed hosts
- ✓ Is message-based – conserves the message boundaries.
- ✓ Classifies messages as:
 - ✓ sequenced delivery of user messages within multiple streams
 - ✓ with an option for un-ordered delivery of individual user messages
- ✓ Additional security mechanisms

LTE / VoLTE IMS : SCTP

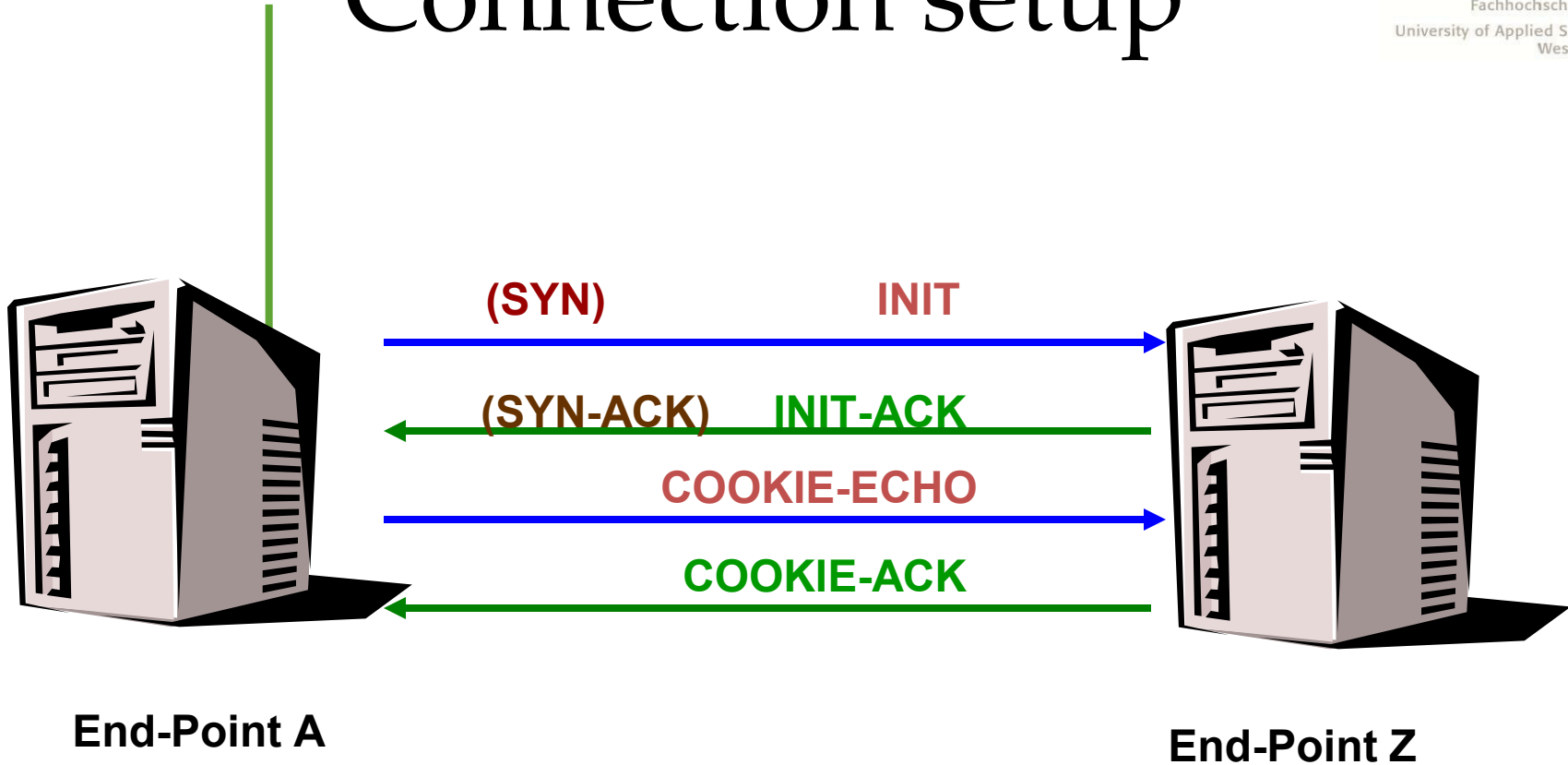


- UDP is message oriented
- TCP is bytes oriented
- SCTP combines the best features of UDP and TCP
- SCTP is a message oriented reliable protocol
- Multistream
- Multihoming

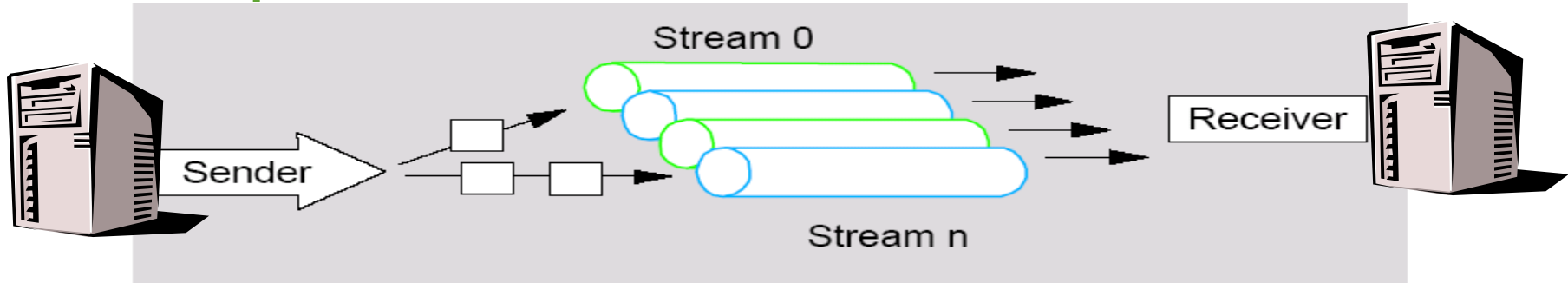
- An association is formed between two endpoints.
- Paths are established between IP addresses owned by those endpoints.



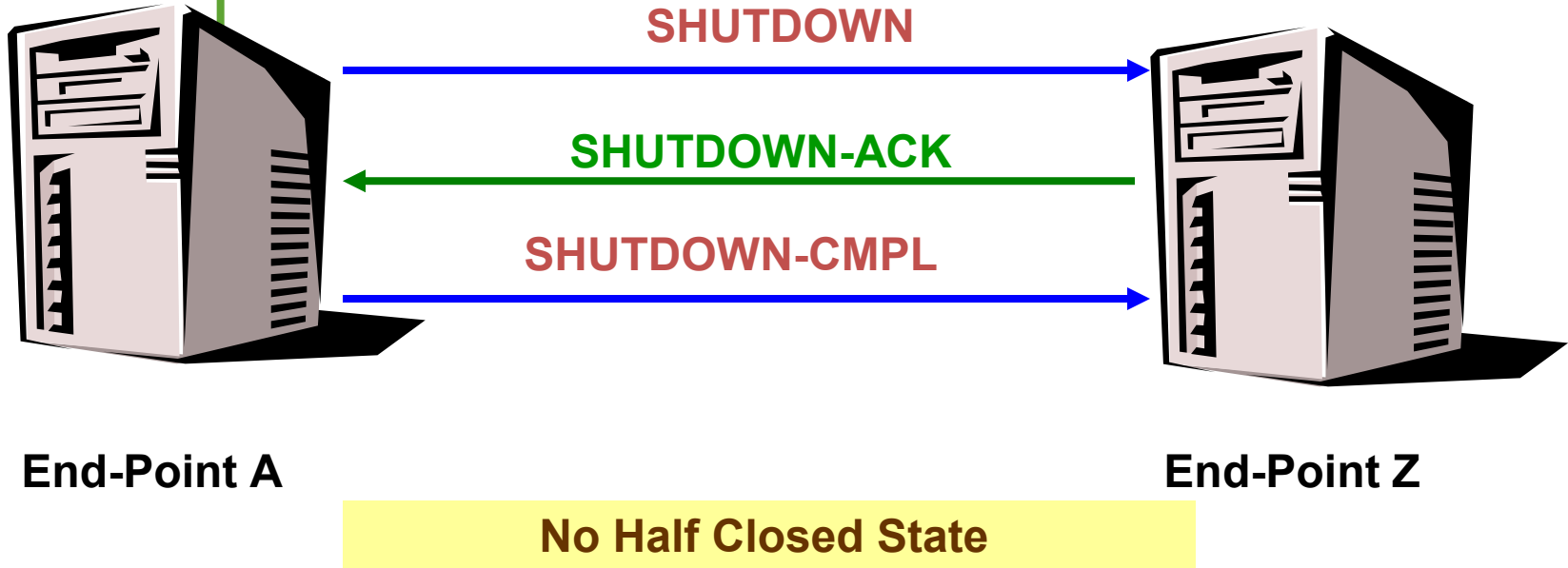
Connection setup



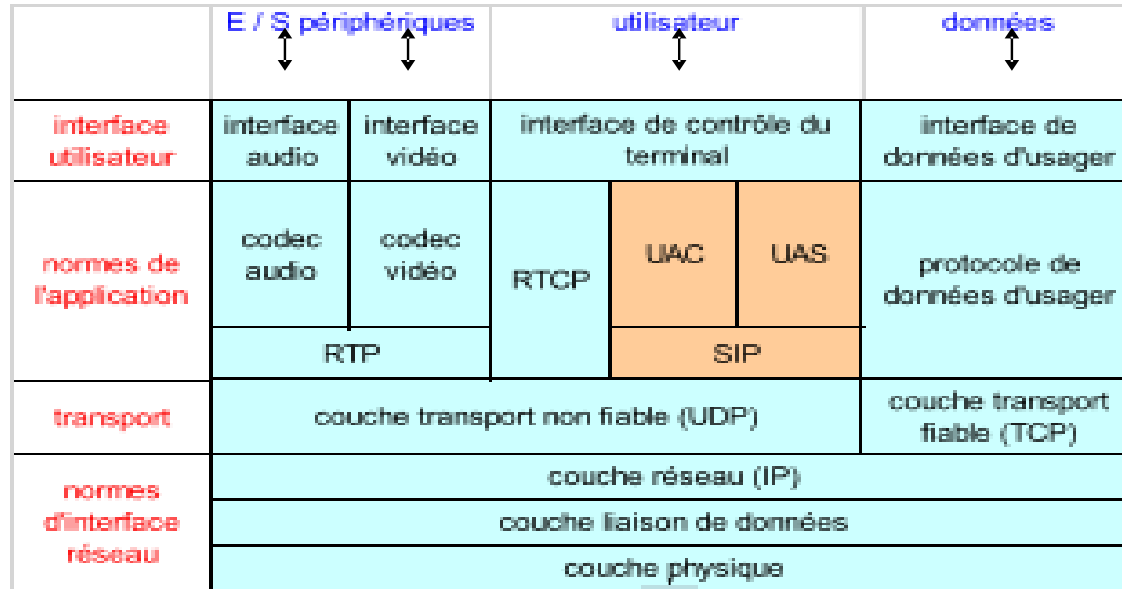
Multi-streaming



Shutdown



Structure normalisée par l'UIT : IETF



UAC : user agent client
UAS : user agent server
SIP : session initiation protocol

LAN, intranet,
internet

Signalizations SIP et H.323

SIP

Text

TCP, UDP, STPC

par lui-même

SDP, simple mais limité

Autres protocoles de l'IETF

Maintien
Transfert d'appel
Mise en attente
Conférence

Messagerie instantanée

H.323

Binaire

TCP

par TCP

H.245, riche mais compliqué

Moyenne

Maintien
Transfert d'appel
Mise en attente
Conférence

Codage

Tansport

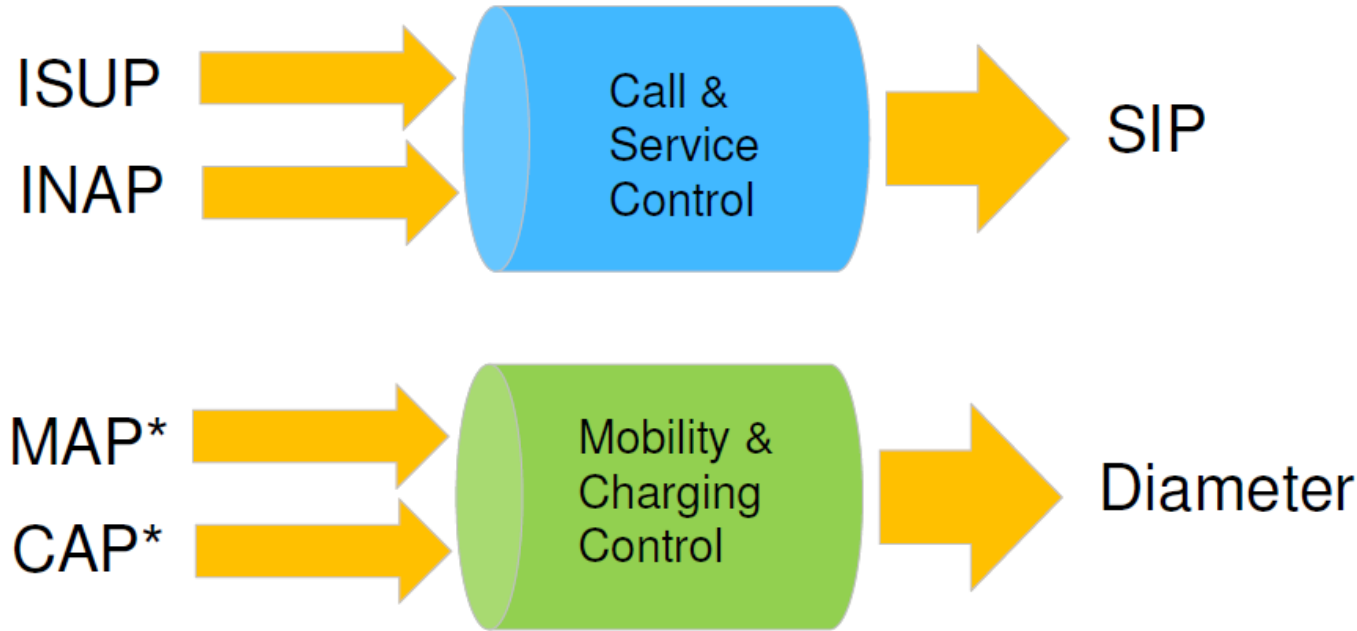
Récupération d'erreurs

Echange des possibilités

Sécurité

Caractéristiques

Diameter & SIP



- › CAP* - 2G/3G CAMEL prepaid functions in future via Diameter, VAS functions of CAMEL via SIP (= INAP)
- › MAP* - AAA and mobility in future via Diameter, Messaging (SMS) via SIP

Diameter vs Radius

	Diameter	RADIUS
Transportation Protocol	Connection-Oriented Protocols (TCP and SCTP)	Connectionless Protocol (UDP)
Security	Hop-to-Hop, End-to-End	Hop-to-Hop
Agent Support	Relay, Proxy, Redirect, Translation	Implicit support, which means the agent behaviors might be implemented in a RADIUS server
Capabilities Negotiation	Negotiate supported applications and security level	Don't support
Peer Discovery	Static configuration and dynamic lookup	Static configuration
Server Initiated Message	Supported. for example, re-authentication message, Session termination	Don't support
Maximum Attribute Data Size	16,777,215 octets	255 octets
Vendor-specific Support	Support both vendor-specific messages and attributes	Support vendor-specific attributes only

Diameter vs RADIUS

Diameter is upwards compatible with RADIUS, so
Messages and AVPs

AVP codes 1-255 is reused from RADIUS

Command codes 0-255 is reused from RADIUS

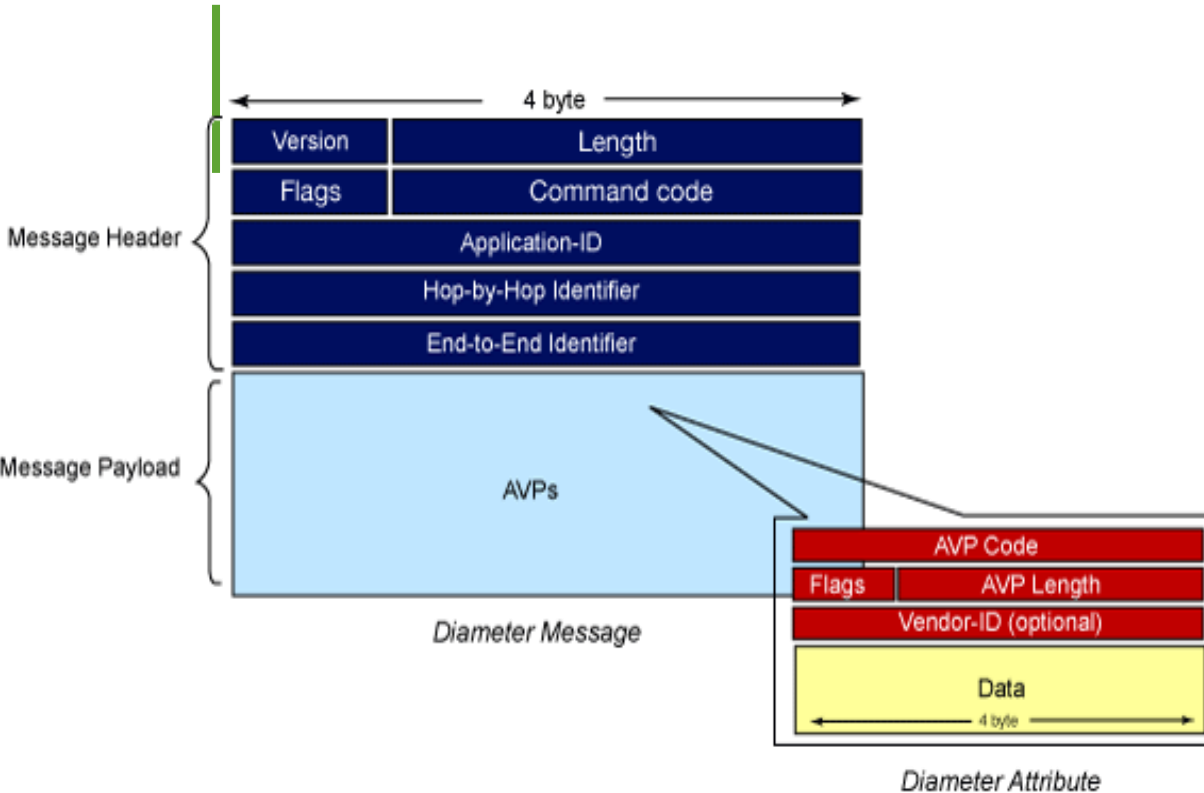
Diameter NASREQ (RFC4005) maps RADIUS
messages to/from Diameter AA-Request and AA-Answer
message

Use of RADIUS<->Diameter Translation Agents

Diameter

- Protocols
 - Certain nodes MUST support at least SCTP or TCP (i.e. Diameter Client)
 - Others MUST support SCTP and TCP (i.e. Diameter Servers and Agents)
- Security
 - TLS and IPSec
- Selection Process (in order of execution)
 - IPSec, SCTP, TCP, TLS
 - SCTP or TCP is always attempted prior to capabilities exchange
 - TLS tried after capability negotiation
 - IPSec and TLS maybe used exclusively

Diameter



Flags

- **"R"** (Request) bit – If set, the message is a request. If cleared, the message is an answer.
- **"P"** (Proxiable) bit – If set, the message MAY be proxied, relayed or redirected.
- **"E"** (Error) bit – If set, the message contains a protocol error.
- **"T"** (Potentially re-transmitted message) bit – This flag is set after a link failover procedure, to aid the removal of duplicate requests.

Cmd AVP

The Result-Code AVP (AVP Code 268) is of type Unsigned32 and indicates whether a particular request was completed successfully or an error occurred. All Diameter answer messages in IETF-defined Diameter application specifications MUST include one Result-Code AVP.

Diameter provides the following classes:

- 1xxx (Informational)
- 2xxx (Success)
- 3xxx (Protocol Errors)
- 4xxx (Transient Failures)
- 5xxx (Permanent Failure)

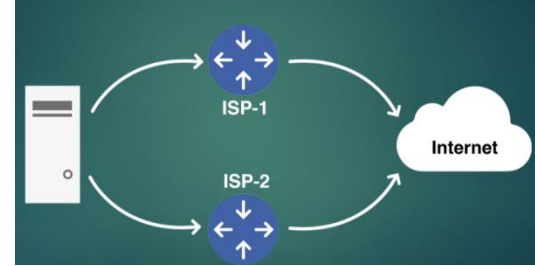
Multimedia-Auth-Answer	MAA	303
Multimedia-Auth-Request	MAR	303
Notify-Answer	NA	323
Notify-Request	NR	323
Profile-Update-Answer	PUA	307
Profile-Update-Request	PUR	307
Push-Notification-Answer	PNA	309
Push-Notification-Request	PNR	309
Push-Profile-Answer	PPA	305
Push-Profile-Request	PPR	305
Re-Auth-Answer	RAA	258
Re-Auth-Request	RAR	258
Registration-Termination-Answer	RTA	304
Registration-Termination-Request	RTR	304
Server-Assignment-Answer	SAA	301
Server-Assignment-Request	SAR	301
Session-Termination-Answer	STA	275
Session-Termination-Request	STR	275
Subscribe-Notifications-Answer	SNA	308
Subscribe-Notifications-Request	SNR	308
Update-Location-Answer	ULA	316
Update-Location-Request	ULR	316
User-Authorization-Answer	UAA	300
User-Authorization-Request	UAR	300
User-Data-Answer	UDA	306
User-Data-Request	UDR	306

Command-Name	Abbr.	Code
AA-Answer	AAA	265
AA-Request	AAR	265
Abort-Session-Answer	ASA	274
Abort-Session-Request	ASR	274
Accounting-Answer	ACA	271
Accounting-Request	ACR	271
Authentication-Information-Answer	AIA	318
Authentication-Information-Request	AIR	318
Bootstrapping-Info-Answer	BIA	310
Bootstrapping-Info-Request	BIR	310
Capabilities-Exchange-Answer	CEA	257
Capabilities-Exchange-Request	CER	257
Credit-Control-Answer	CCA	272
Credit-Control-Request	CCR	272
Device-Watchdog-Answer	DWA	280
Device-Watchdog-Request	DWR	280
Diameter-EAP-Answer	DEA	268
Diameter-EAP-Request	DER	268
Disconnect-Peer-Answer	DPA	282
Disconnect-Peer-Request	DPR	282
Location-Info-Answer	LIA	302
Location-Info-Request	LIR	302
Message-Process-Answer	MPA	311
Message-Process-Request	MPR	311

Errors list

Value	Message Name	Description
3001	DIAMETER_COMMAND_UNSUPPORTED	The Request contained a Command-Code that the receiver did not recognize or support.
3002	DIAMETER_UNABLE_TO_DELIVER	This error is given when Diameter can not deliver the message to the destination.
3003	DIAMETER_REALM_NOT_SERVED	The intended realm of the request is not recognized.
3004	DIAMETER_TOO_BUSY	When returned, a Diameter node should attempt to send the message to an alternate peer.
3005	DIAMETER_LOOP_DETECTED	An agent detected a loop while trying to get the message to the intended recipient.
3006	DIAMETER_REDIRECT_INDICATION	A redirect agent has determined that the request could not be satisfied locally.
3007	DIAMETER_APPLICATION_UNSUPPORTED	A request was sent for an application that is not supported.
3008	DIAMETER_INVALID_HDR_BITS	A request was received whose bits in the Diameter header were either set to an invalid combination.
3009	DIAMETER_INVALID_AVP_BITS	A request was received that included an AVP whose flag bits are set to an unrecognized value.
3010	DIAMETER_UNKNOWN_PEER	A CER was received from an unknown peer.

MPTCP

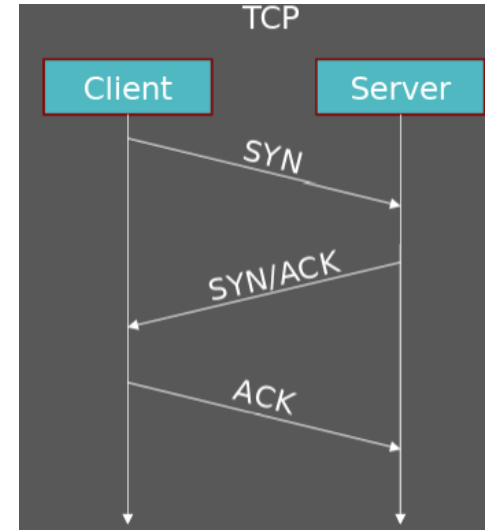
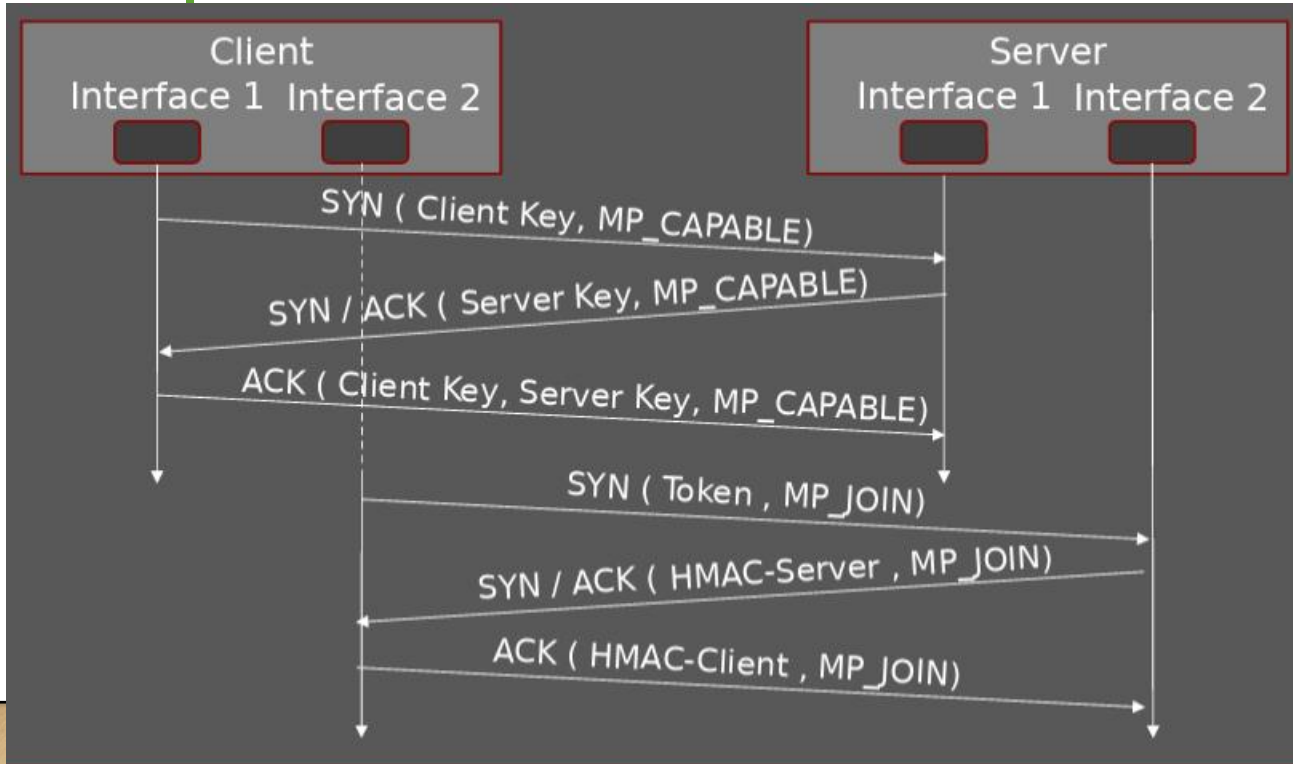


Transferts transparents : passage d'un chemin à un autre tout en préservant les connexions établies, par exemple Apple utilise le protocole TCP Multipath sur les smartphones.

Sélection du meilleur réseau : utilisation du « meilleur » chemin disponible en fonction de certaines conditions, par exemple la latence, les pertes, le coût, la bande passante, etc.

Agrégation de réseaux : utilisation simultanée de plusieurs chemins pour obtenir un débit plus élevé, par exemple pour combiner des réseaux fixes et mobiles afin d'envoyer des fichiers plus rapidement.

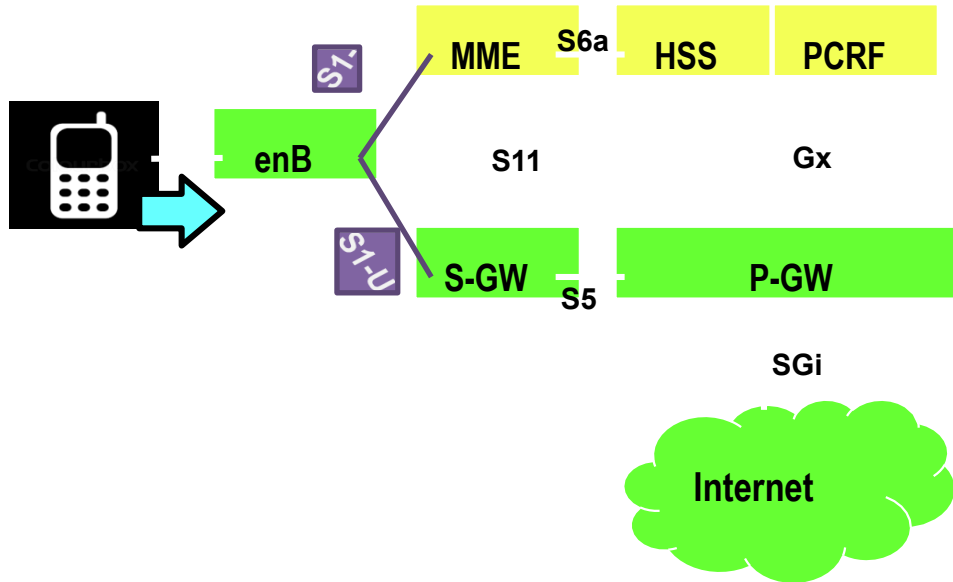
TCP vs MPTCP



PAQUET MPTCP

Bits 0-7		Bits 8-15		Bits 16-23		Bits 24-31	
Source Port				Destination Port			
TCP Sequence Number							
TCP Acknowledge Number							
Data Offset	Reserved	TCP Flags (Ack, Syn etc.)			Window Size		
Checksum				Urgent Pointer			
MPTCP Option Type		Length		Subtype	MPTCP Vers.	MPTCP Flags	
Remaining MPTCP Subtype Data							
Payload							

LTE to VoLTE Journey



eNB (EnodeB)

MME (Mobility Management Entity)

SGW (Serving Gateway)

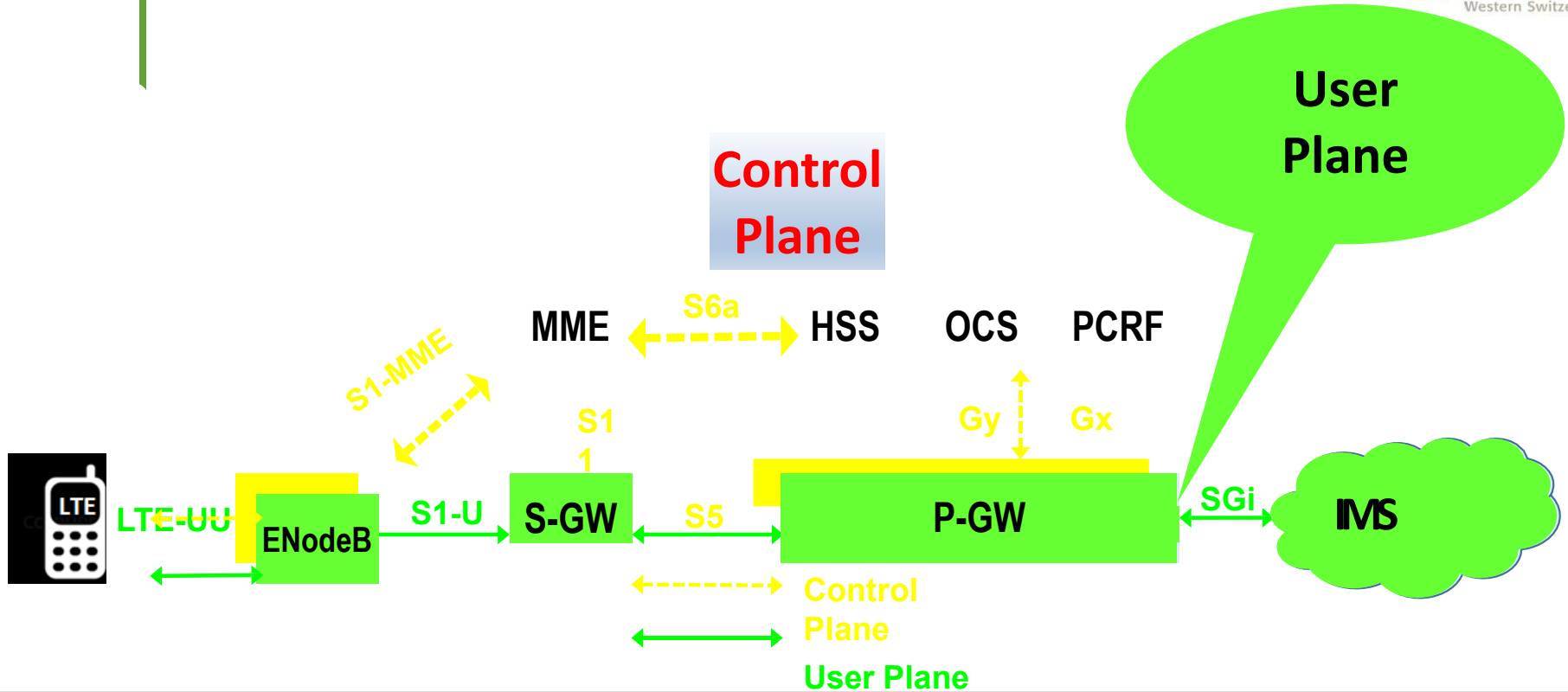
HSS (Home Subscriber Server)

PCRF (Policy charging rules function)

PGW (Packet Gateway)

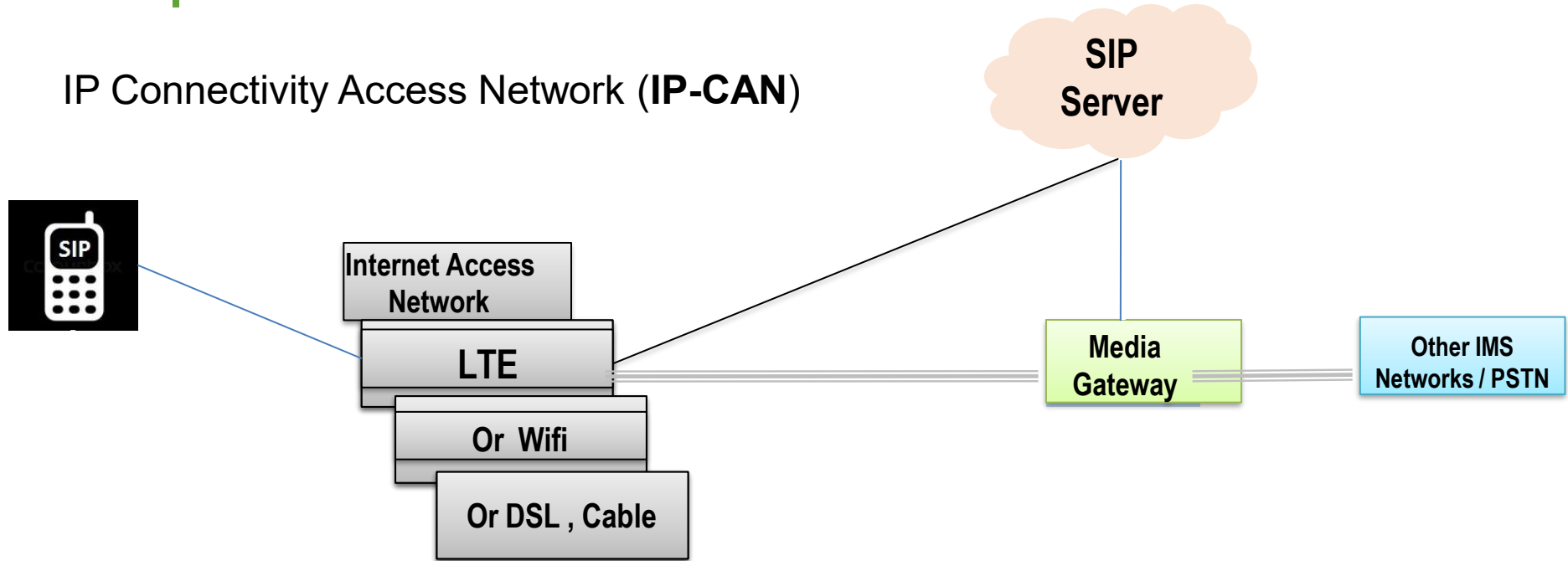
OCS (Online Charging System)

LTE Architecture

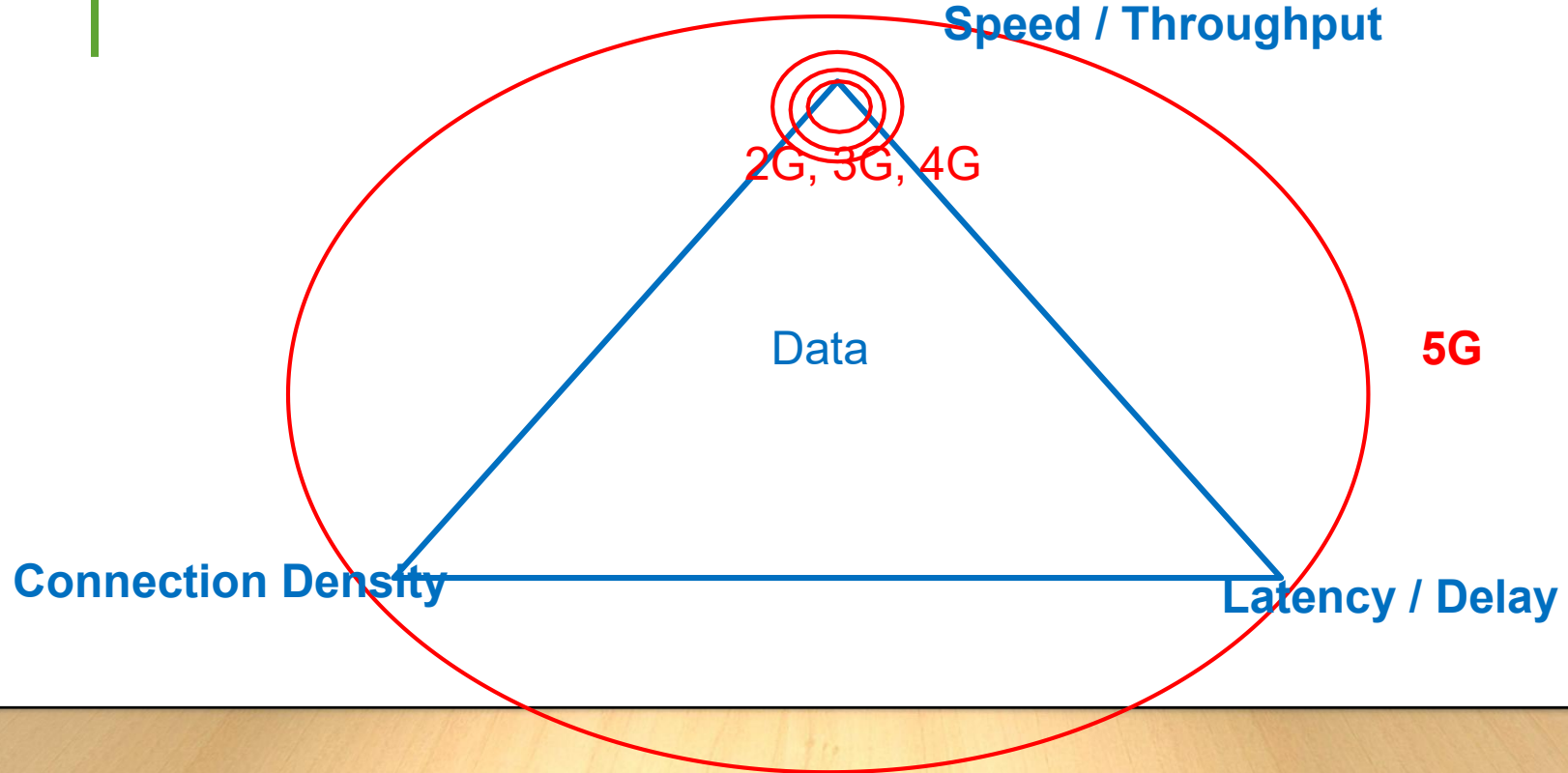


Role of IPCAN

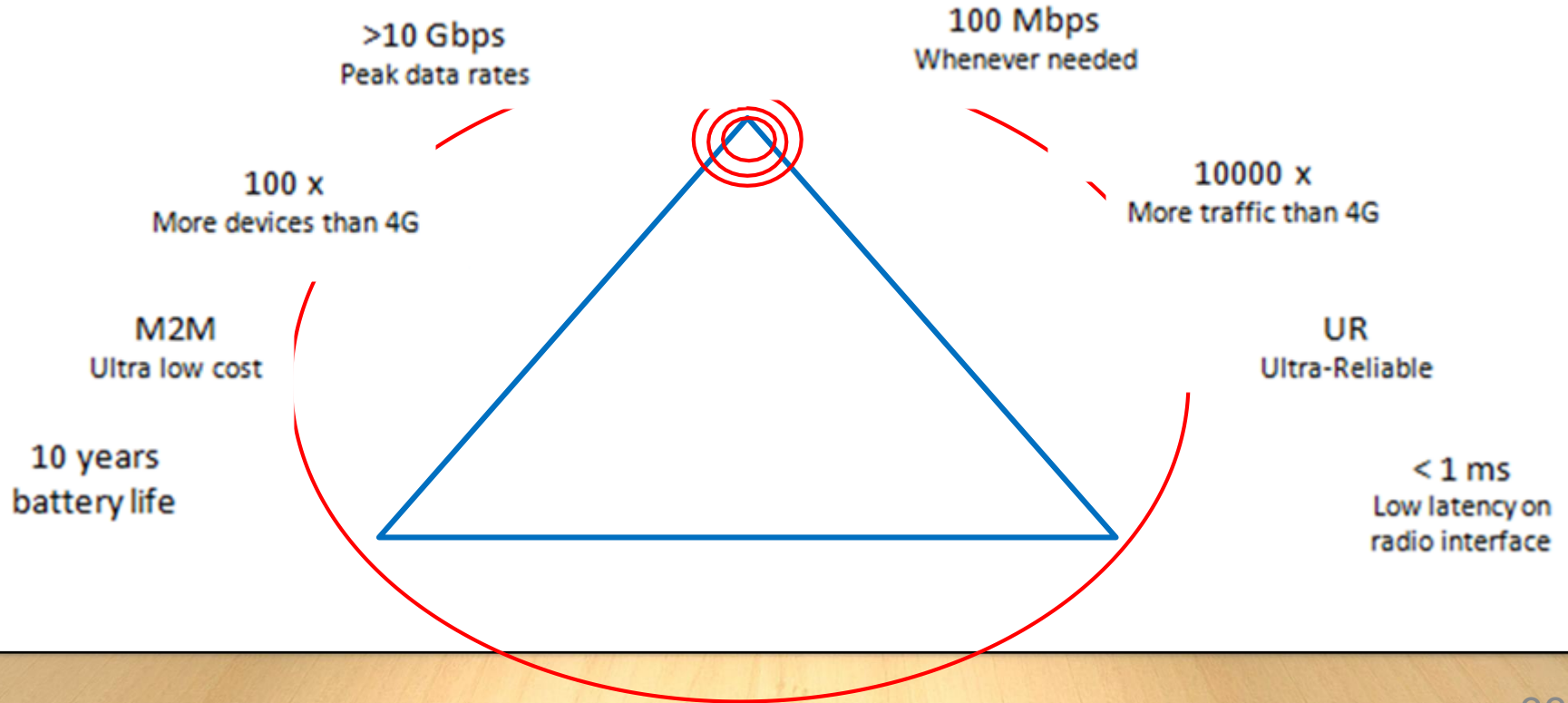
IP Connectivity Access Network (IP-CAN)



Les évolutions

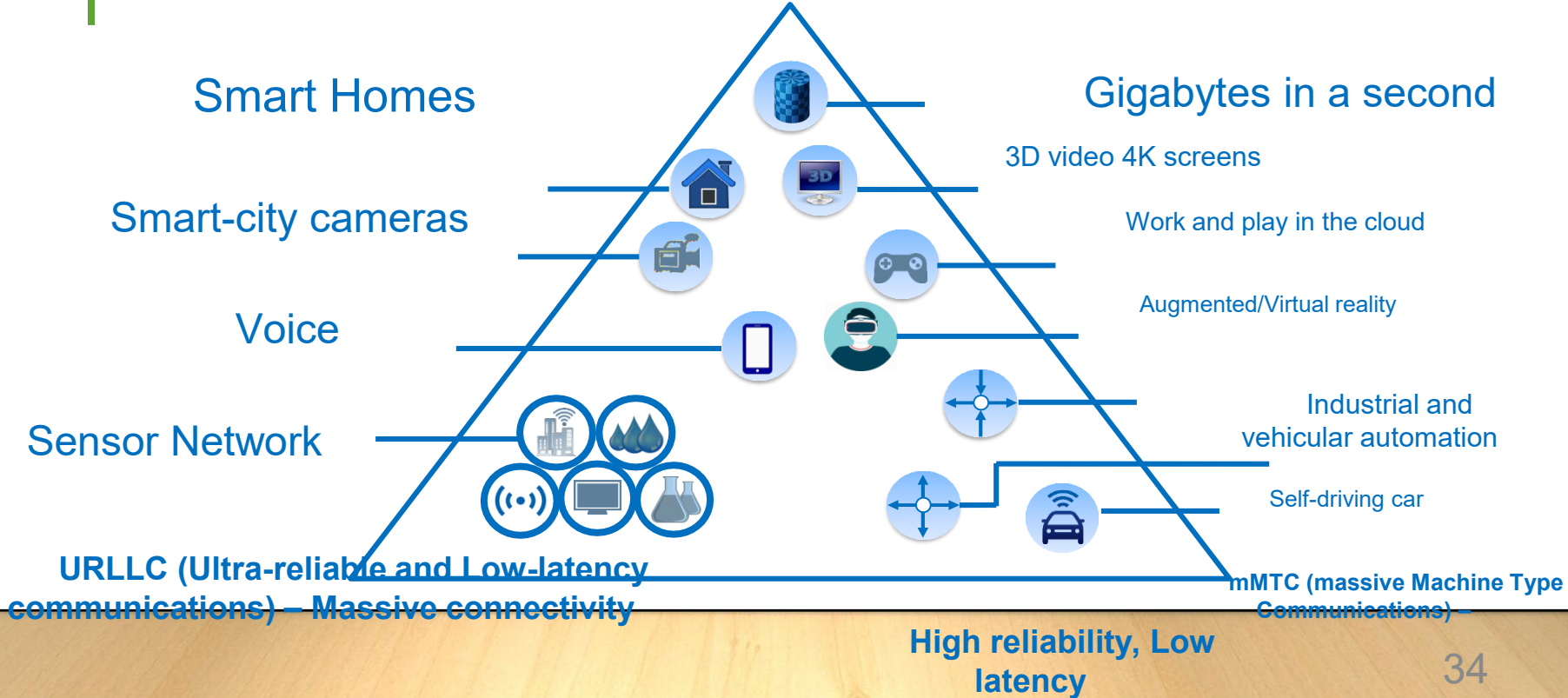


Les objectifs 5G



5G

eMBB (enhanced Mobile Broadband) – Capacity Enhancement



Les contraintes Economiques

Time to Market

Economy of Scale

Economy of Scope

Energy optimization

Autonomy

User Centric

User Experience

Ubiquitous

Contraintes techniques

Couverture globale

Convergence IP

Convergence Fixe_Mobile

Zero_Trust Security

Cross-Layring Security

Couverture Globale

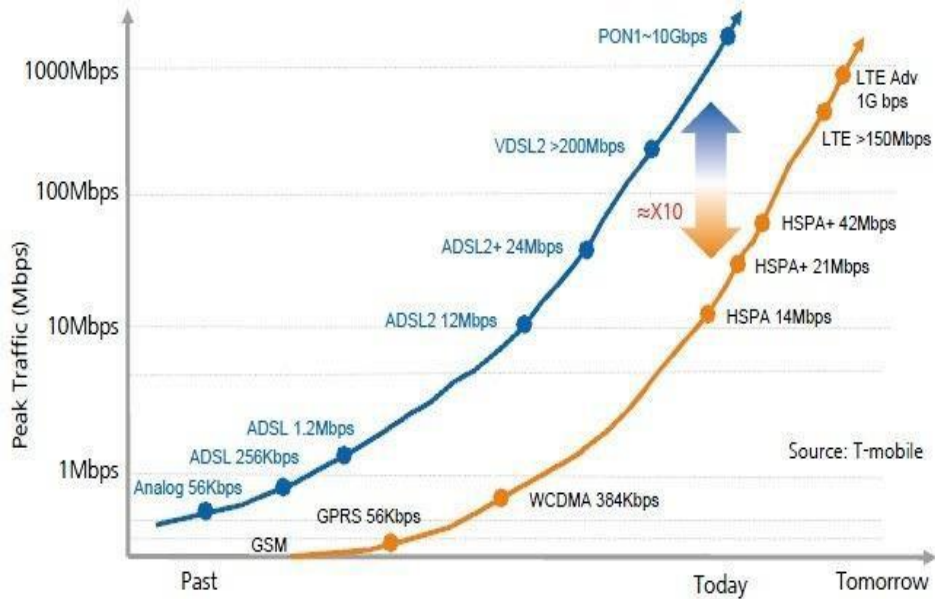
Latence et Gigue

Débit

SDN

Réseaux Privés Open-RAN

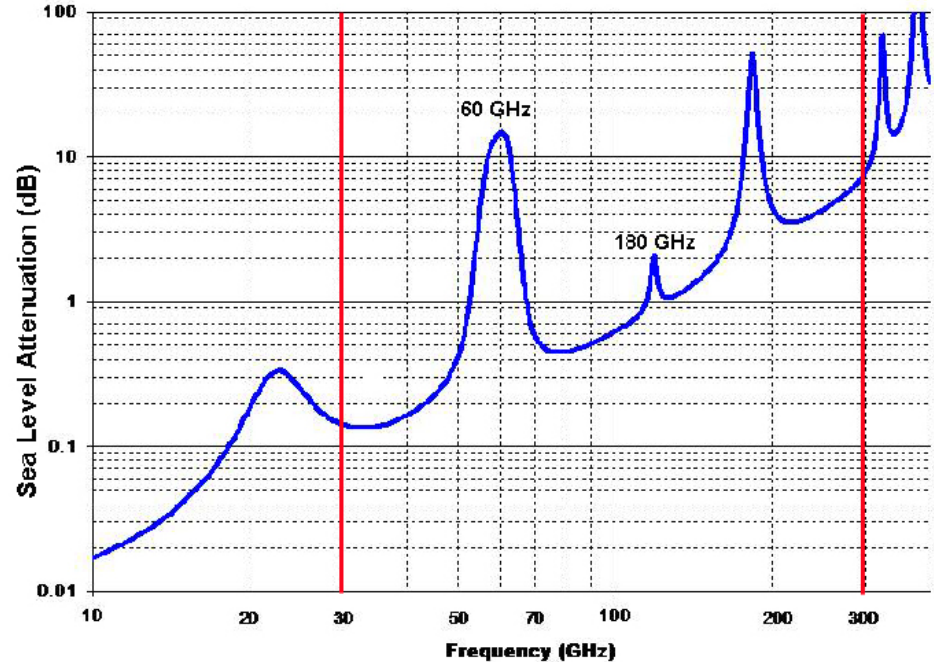
Mobile/Wi-Fi



IEEE Standard	Year Adopted	Frequency	Max. Data Rate	Max. Range
802.11a	1999	5 GHz	54 Mbps	400 ft.
802.11b	1999	2.4 GHz	11 Mbps	450 ft.
802.11g	2003	2.4 GHz	54 Mbps	450 ft.
802.11n	2009	2.4/5 GHz	600 Mbps	825 ft.
802.11ac	2014	5 GHz	1 Gbps	1,000 ft.
802.11ac Wave 2	2015	5 GHz	3.47 Gbps	10 m.
802.11ad	2016	60 GHz	7 Gbps	30 ft.
802.11af	2014	2.4/5 GHz	26.7 Mbps – 568.9 Mbps (depending on channel)	1,000 m.
802.11ah	2016	2.4/5 GHz	347 Mbps	1,000 m.
802.11ax	2019	2.4/5 GHz	10 Gbps	1,000 ft.
802.11ay	late 2019	60 GHz	100 Gbps	300-500 m.
802.11az	2021	60 GHz	Device tracking refresh rate 0.1-0.5 Hz	Accuracy <1m to <0.1m

Différents Affaiblissements

Affaiblissement par le Trajet
Affaiblissement par absorption
Affaiblissement dans l'atmosphère
Affaiblissement due à l'eau
Affaiblissement par diffraction
Evanouissement = fading
(multi-trajet)



Affaiblissement

$$\text{LOS } A = 32 + 20 \text{ Log}(F_{\text{MHZ}}) + 20 \text{ Log}(d_{\text{km}})$$

$$\text{NLOS } A = 32 + a \cdot 20 \text{ Log}(F_{\text{MHZ}}) + b \cdot 20 \text{ Log}(d_{\text{km}})$$

$$1 < a < 2 \text{ et } 1 < b < 3$$

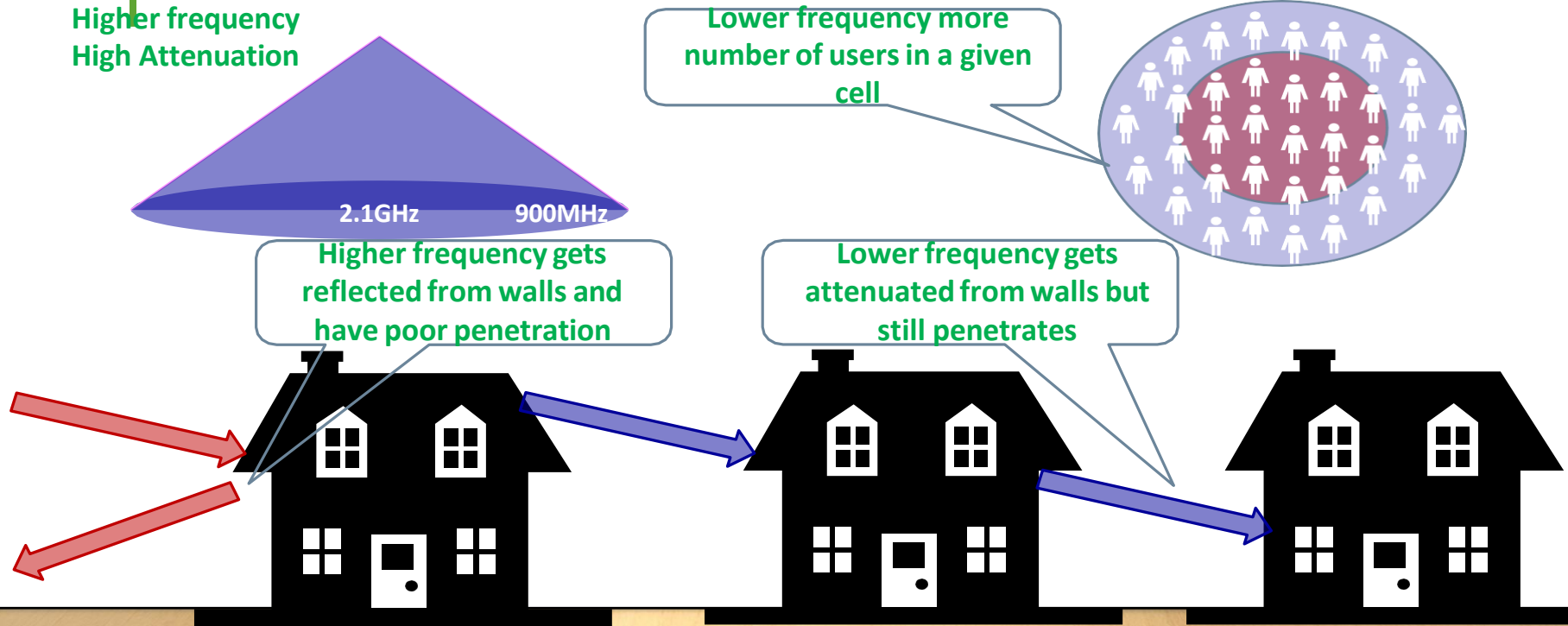
Puissance de 1W (30dBm) à 1mw (0dBm),

$P_{\text{station}} = 10\text{W}$ à 300W

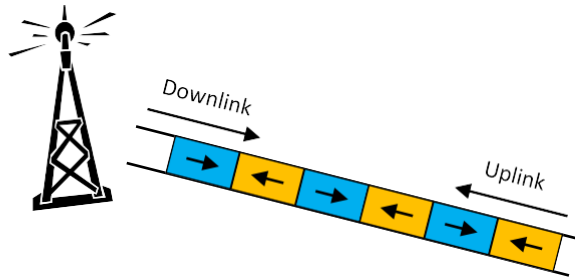
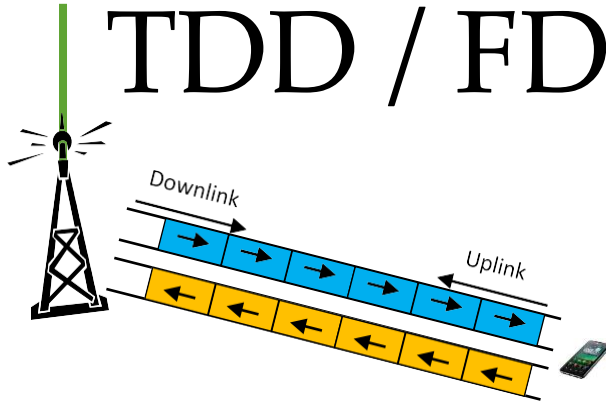
Batterie 20Ah à 40 Ah

Gain d'antenne 3 dBi; 20 dBi

Frequency selection



TDD / FDD



Frequency Division Duplex (FDD)

Simpler to implement

Simultaneous downlink and uplink trans

No need for synchronisation hence simp

Needs paired spectrum UL/DL ratio is fix

Time Division Duplex (TDD)

Implementation is complex

Only uplink (UL) or downlink (DL) at any

Need for synchronisation within the who

No need for paired spectrum

Number of UL/DL ratio is changeable

Latency & Jitter -délai et gigue

Latency is generally defined as the time it takes for a source to send a packet of data to a receiver. In simple terms, half of Ping time. This is also referred to as **one way latency**.

Sometimes the term **Round trip latency** or round trip time (RTT) is also used to define latency. This is the same as ping time.

Jitter is defined as the variation in the delay (or latency) of received packets. 'delay jitter'.

Evolutions

1G : Analogique

2G : Numérique

3G : Data

4G : Débit

5G : Latence

6G : Haute fréquence

7G :

Low latency

End-to-end (E2E) Latency : the time that takes to transfer a given piece of information from a source to a destination, measured at the communication interface, from the moment it is transmitted by the source to the moment it is successfully received at the destination.

1G: Latence moins pertinente

2G : 300-600ms

3G : 100-500 ms

4G : 50-100 ms LTE-A 20ms

5G : 1-10ms

What if LTE Radio is Congested ??

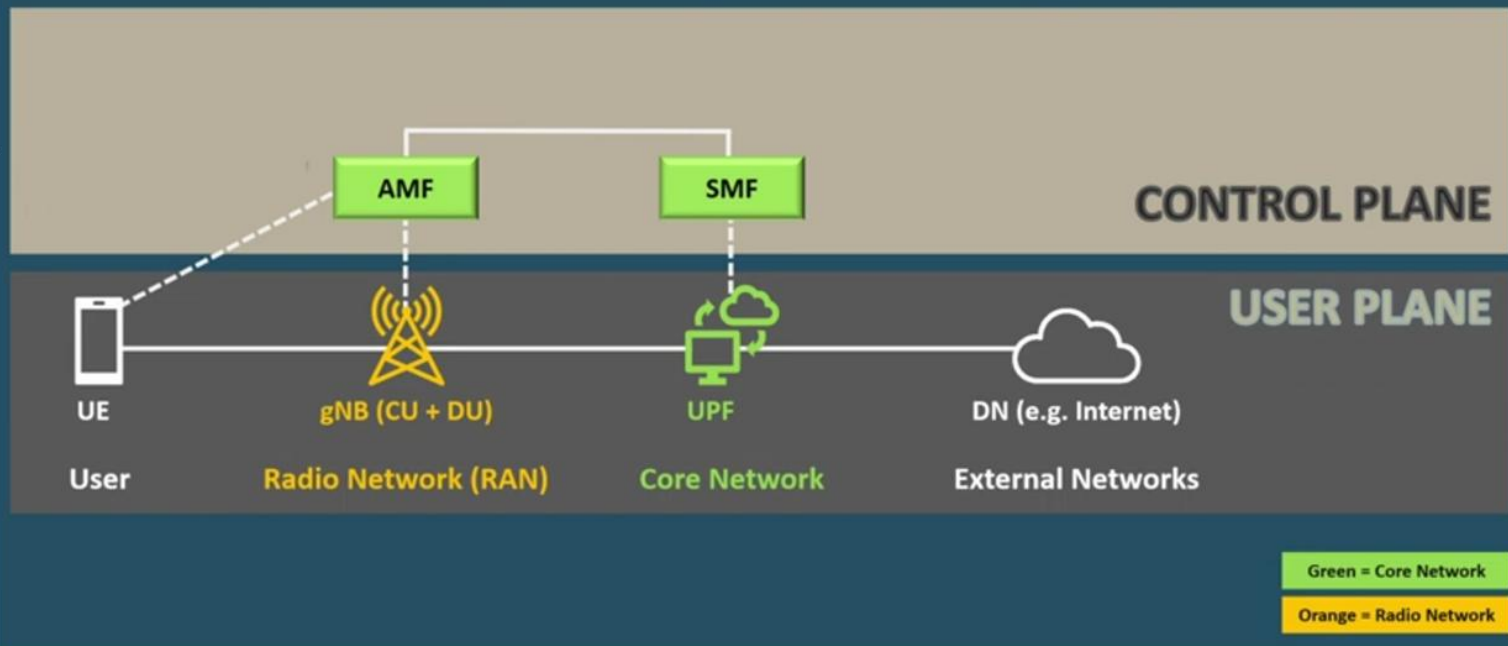
Priority	QCI	Resource Type	Packet Delay Budget	Packet Error Loss Rate	Example Services
1	5	non-GBR	100ms	10^{-6}	Volte , IMS Signaling
2	1	Guaranteed Bit Rate (GBR)	100ms	10^{-2}	Volte , Conversational Voice
3	3		50ms	10^{-3}	Real Time Gaming, V2X messages
4	2		150ms	10^{-3}	Volte , Conversational Video (Live Streaming)
5	4		300ms	10^{-6}	Non-Conversational Video (Buffered Streaming)
6	6	non-GBR Best Effort	300ms	10^{-6}	Internet Services , Video (Buffered Streaming) TCP-Based (for example, www, email, chat, ftp, p2p and the like)
7	7		100ms	10^{-3}	
8	8		300ms	10^{-6}	
9	9		300ms	10^{-6}	

5G SA vs NSA

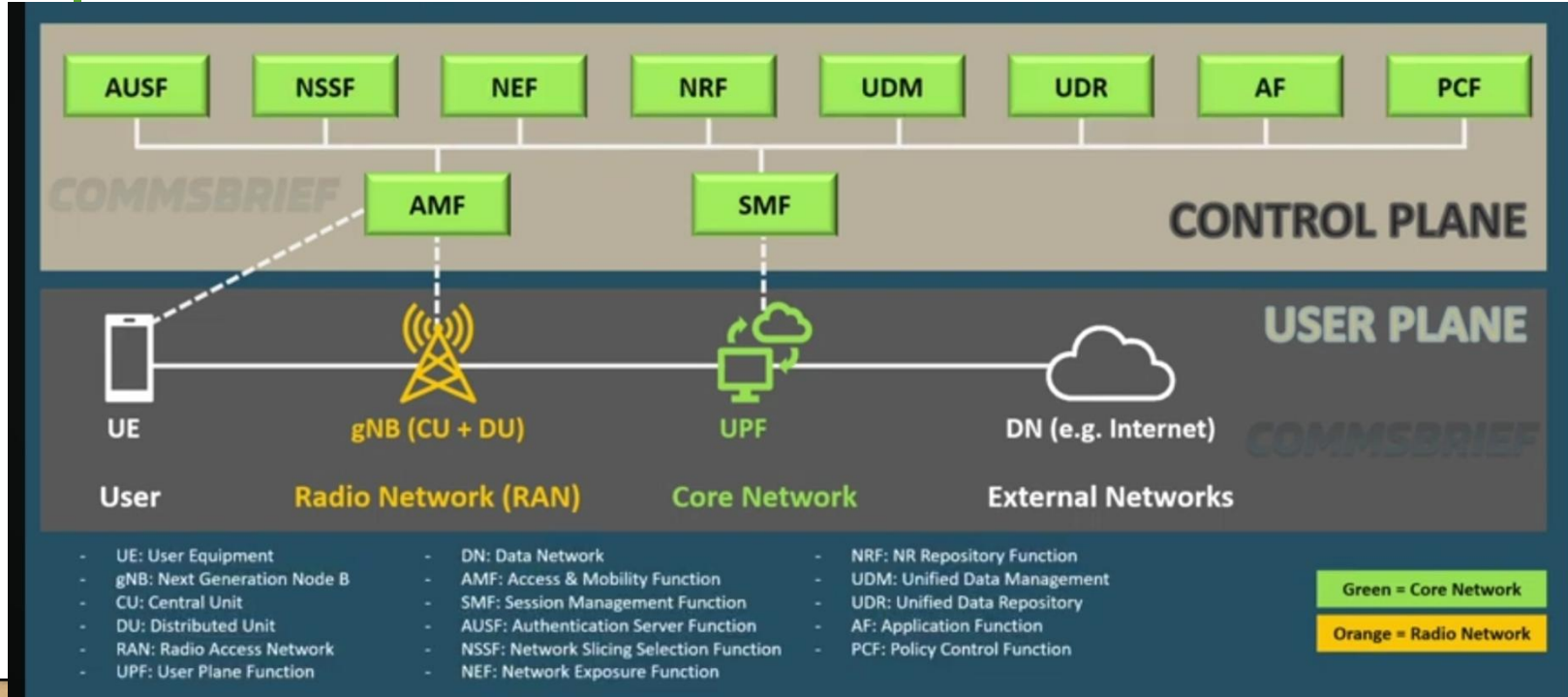
5G Standalone (5G SA) is a 5G network operating independently of 4G, using a dedicated 5G core (5GC) instead of relying on older infrastructure. It delivers on the full potential of 5G by offering significantly faster speeds, ultra-low latency, and massive IoT connectivity.

NSA is a collaborating 4G and 5G

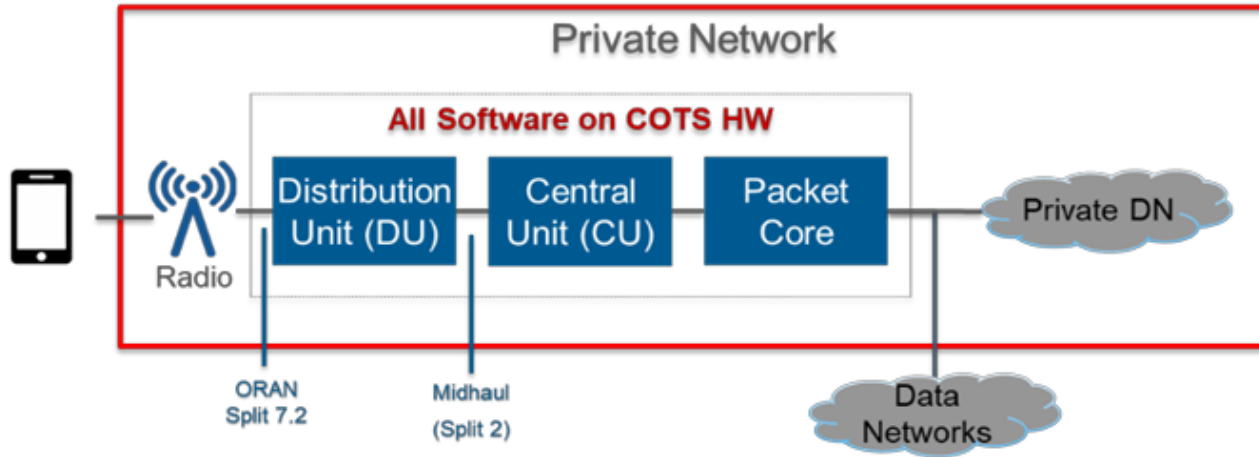
5G Stand Alone Network



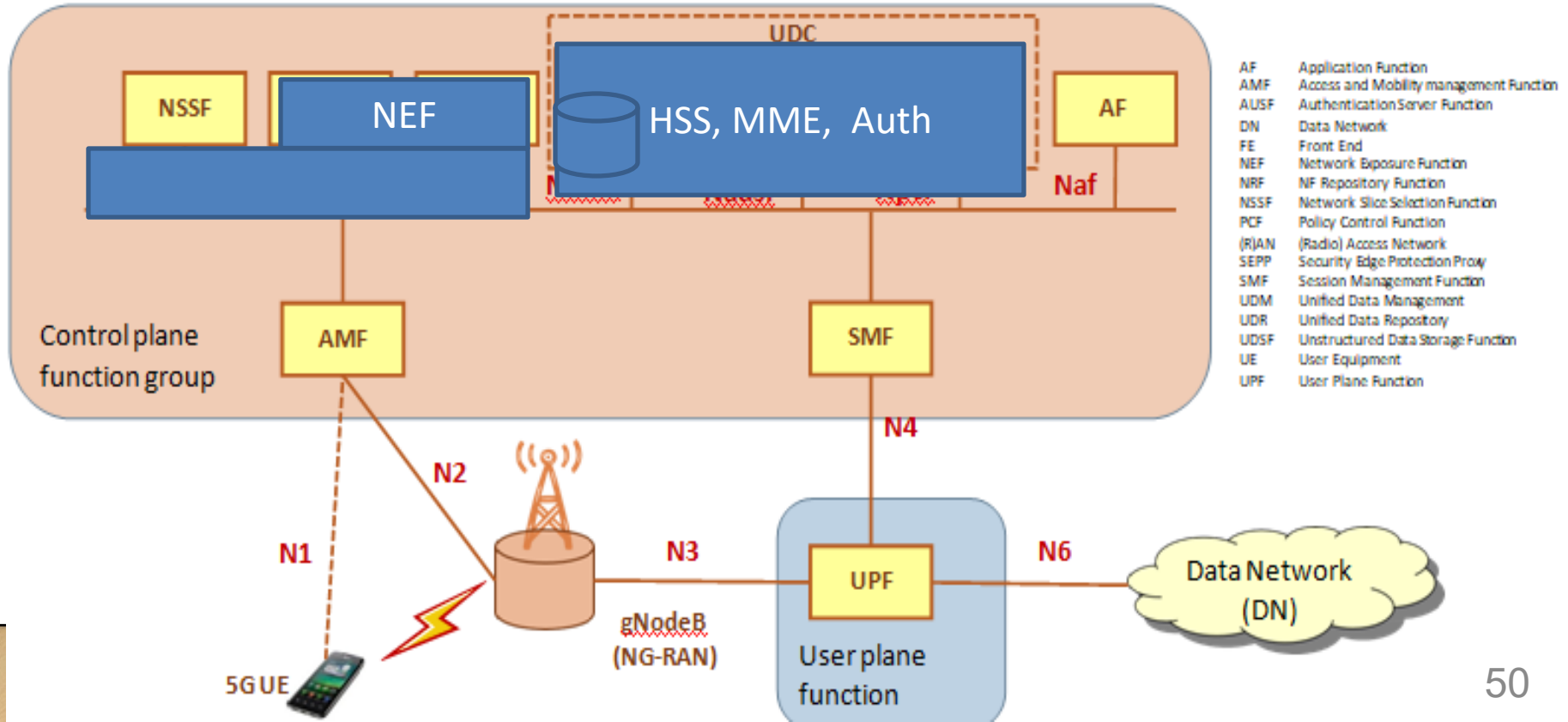
5G stand alone Network



5G



5G architecture



Terminologie

Unified Data Management (UDM), is analogous to the Home Subscriber Server (HSS) in EPC architecture and introduces the concept of User Data Convergence (UDC)

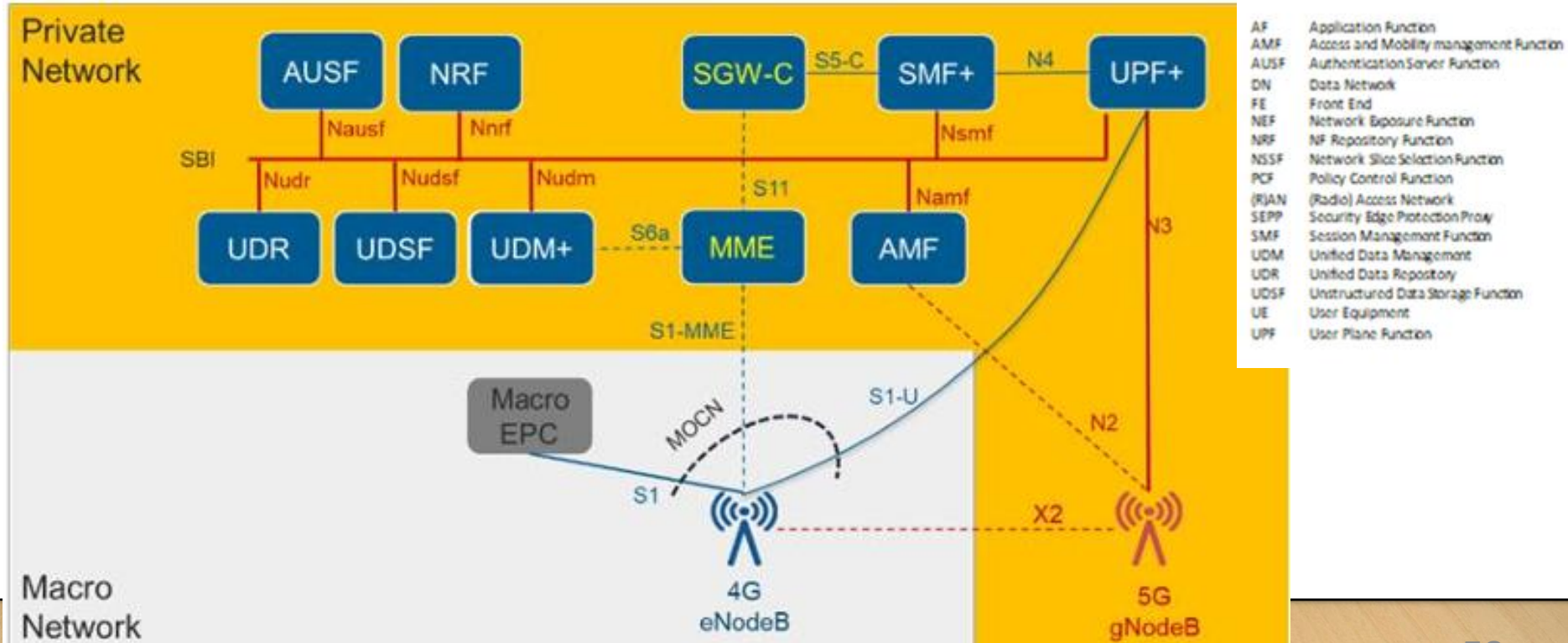
Authentication Server Function (AUSF) AUC

SMF : Roaming functionality, Handle local enforcement to apply QoS SLAs (VPLMN), Charging data collection and charging interface (VPLMN), Lawful intercept

SAF Registration management, Connection management.

Reachability management, Mobility Management.

Fonctions



References

- GSMA FCM.01
- GSMA IR.92 (Résumé)
- IMS SPECS 3GPP 23.228
- 5G: An Advanced : ZAHID GHADIALY